



# Physical informed neural network for SOH estimation of lithium-ion battery with electrochemical mechanism

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## ABSTRACT

In recent years, Physics-Informed Neural Networks (PINNs) have emerged as a promising research approach for estimating the State of Health (SOH) of lithium-ion batteries. Compared to traditional purely data-driven methods, PINNs embed physical laws directly into the neural network's loss function. This integration enhances predictive accuracy and generalization while ensuring physical consistency, thereby improving the model's interpretability. Therefore, this study proposes a EM-PINN model based on the electrochemical degradation mechanism of solid electrolyte interphase (SEI) layer growth. First, the battery capacity degradation process is modeled from the perspective of the electrochemical principles governing SEI layer growth, establishing explicit correlations between SEI layer growth kinetics and the rate of SOH degradation to form the physical constraints. Second, the EM-PINN framework is designed, incorporating the partial differential equation (PDE) loss, boundary condition loss, initial condition loss, and SOH physical loss to jointly guide the learning process of the model, thereby enhancing both its interpretability and prediction accuracy. Experimental validation on three public datasets (XJTU, MIT, and HUST) demonstrates that the proposed EM-PINN method achieves high-accuracy SOH estimation across different datasets. Notably, across six testing scenarios with different data volumes, the mean absolute percentage error (MAPE) reaches as low as 0.873%. This study provides a novel solution for integrating electrochemical mechanisms with PINN models for battery SOH estimation.

## 1. Introduction

The accelerating global energy transition [1] and pursuit of carbon neutrality [2] goals are profoundly reshaping transportation and energy systems. Electric vehicles (EVs), serving as the pivotal solution for greening the transportation sector, are being deployed on a large scale. Concurrently, Vehicle to Grid (V2G) technology [3,4] is emerging as a critical component for achieving deep synergy between the transportation and energy sectors. This innovative approach enables bidirectional energy flow, not only drawing power from the grid but also feeding stored energy from EV batteries back into it.

Lithium-ion batteries (LIBs) serve as fundamental energy storage units in EVs and V2G systems [5,6]. Their performance directly affect the driving range, safety, and service life of the vehicle, as well as the economic benefits and reliability of V2G participation. However, battery aging is an inevitable process. The SOH of the battery progressively degrades to failure under the influence of factors such as usage time [7], charge/discharge cycles [8], depth of discharge (DOD) [9], C-rate [10], and temperature [11]. Consequently, ensuring the safe, reliable, and

efficient operation of LIBs requires long-term condition monitoring (including SOH assessment [12]) and appropriate maintenance, both of which are crucial. As is well known, the SOH is quantified by the ratio of available capacity to initial capacity, and is employed as an essential indicator for assessing battery degradation [13].

Currently, SOH estimation methods for LIBs can be broadly categorized into three types: physics-based models [14,15], data-driven [16–18], and hybrid methods [19,20]. Among them, physics-based models typically employ multiphysics-coupled approaches to simulate internal degradation mechanisms and quantify battery degradation. For instance, Tang et al. [21] developed an electrochemical–thermal model incorporating loss of lithium inventory (LLI) and SEI layer growth to predict LIB capacity degradation. However, to more accurately characterize the battery degradation process, Luo et al. [22] proposed an electrochemical–thermal–mechanical coupled model to characterize capacity degradation. This framework incorporates stress effects on degradation behavior and introduces temperature-dependent dynamic

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parameters for degradation assessment. Meanwhile, Li et al. [23] developed a single-particle model (SPM) coupling chemical and mechanical degradation mechanisms, which enables rapid prediction of capacity degradation and voltage curve evolution with a prediction error of approximately 1%.

Although physics-based models can accurately describe the electrochemical behavior during battery degradation and achieve high prediction accuracy, the LIB aging process inherently couples multiphysics domains such as electrochemistry, thermal effects, and mechanics. This inherent multiphysics coupling significantly increases model complexity and the computational cost of solving multi-scale models. Moreover, solving these intricate models demands substantial resources, thus compromising real-time capability and limiting their practicality for online applications.

To address these challenges, data-driven approaches have attracted growing attention from researchers [24]. For instance, Deng et al. [25] proposed a data-driven approach using random partial charging processes with sparse Gaussian process regression. This method extracts features from local charging segments for SOH estimation, achieving mean maximum absolute errors of 2.88%, 2.52%, and 1.51% across three validation cases. Furthermore, in order to combine the feature information contained in electrochemical impedance spectra (EIS) with deep neural network methods, Luo et al. [26] designed a transformer-based model for automated feature extraction from EIS data, reducing the MAPE to 1.63% for battery SOH estimation. Meanwhile, in an effort to reduce model complexity, Wu et al. successively proposed a lightweight neural network based on deformable depthwise convolution [27] and a lightweight deformable network based on knowledge distillation [28] (KD-LDNet) for predicting the remaining useful life of lithium-ion batteries. Both approaches ultimately enhanced the level of lightweight design while maintaining high predictive accuracy. Although the aforementioned data-driven methods possess strong nonlinear fitting capability and high efficiency, their reliance on data and selected features significantly limits generalization performance [29]. Moreover, as these approaches rely entirely on data for outcome prediction, the models lack physical interpretability to a certain extent.

The rapid development of deep learning algorithms [30] has driven the advancement of hybrid models that combine physical models with data-driven methods in the field of LIB SOH estimation. These models integrate the strengths of both paradigms, overcoming the limitations of traditional single physical or purely data-driven methods, while maintaining high prediction accuracy and enhancing model interpretability. For example, Ye et al. [31] proposed a Physics-Informed Feed-forward Neural Network (PIFNN), which incorporates the monotonic relationship between P-IC and SOH as physical constraints into the neural network's loss function to guide model training. During prediction, a physics-constrained secondary training process is applied to the FNN predictions, reducing the prediction error to below 1.5%. Wen et al. [32] constructed a PINN framework based on semi-empirical PDEs, utilizing a data-driven Deep Hidden Physics Model (DEEP-HPM) to mine inherent governing equations when prior knowledge of dynamics is scarce, and fusing the obtained dynamic information with the outputs of the PINN surrogate networks to predict SOH. Wang et al. [33] proposed a PINN framework integrating empirical models with state-space equations. This approach employs neural networks to approximate the dynamic process of battery degradation and incorporates physical constraints from generalized PDE equations into the loss function. Experimental results demonstrate that the framework achieves a MAPE below 0.87% across four large-scale datasets.

However, in most existing PINN frameworks, physical constraints are generally constructed by incorporating semi-empirical or empirical models into the loss function of the neural network. This approach fails to effectively integrate physical constraints derived from multi-physics models that are more consistent with physical laws, thereby limiting the

PINN model's capability to accurately describe battery aging processes and represent the underlying physical mechanisms.

To address the aforementioned issues, this study focuses on the electrochemical process of SEI layer growth within LIBs, which is recognized as a key mechanism responsible for battery capacity fade [34, 35]. SEI layer growth primarily originates from the irreversible reduction reaction of electrolyte solvents at the anode surface [36], which leads to continuous SEI layer thickening and irreversible consumption of active lithium ions [37]. Concurrently, this growth reduces the porosity within the SEI layer [38], consequently increasing the impedance to lithium-ion diffusion and ultimately resulting in capacity fade. Thus, it can be inferred that an inseparable relationship exists between the growth of the SEI layer and the decline in battery SOH.

This study aims to integrate the physical information from this capacity degradation mechanism, directly caused by SEI layer growth, into neural network modeling. Based on this mechanism, a PINN framework is developed for precise SOH estimation. In contrast to conventional PINN approaches reported in the literature, this study not only fully accounts for the impact of SEI layer growth on battery aging but also constructs an explicit and known mathematical expression, thereby enabling a more accurate characterization of the battery degradation process. Specifically, the explicit relationship between SEI film growth and SOH decay is transformed into physical knowledge, and the corresponding physical constraints are embedded into the training process of the neural network in the form of loss functions, to guide its learning. This enhances the model's interpretability while improving the accuracy and stability of SOH prediction. The proposed approach not only ensures reliable and precise SOH estimation but also provides novel perspectives and methodologies for integrating electrochemical models within PINN frameworks. This study makes the following key contributions:

- (1) Based on the coupled electrochemical degradation mechanisms involving SEI layer growth, a physics-based model was developed to characterize battery degradation. This model establishes an explicit link between SEI layer growth and SOH degradation, which is subsequently transformed into physical constraints to guide the learning process of the subsequent model.
- (2) An EM-PINN framework is proposed. The framework integrates multiple physical constraints, including PDE loss, boundary condition loss, initial condition loss, and SOH physical loss, into the neural network's training, thereby improving the model's prediction stability and accuracy.
- (3) The performance of the proposed model was validated on both the XJTU, MIT and HUST datasets, with discussions focusing on its accuracy and stability. Experimental results demonstrate that the EM-PINN model maintains high estimation accuracy, even when learning from small-sample datasets.

The structure of this paper is organized as follows. The [Introduction section](#) presents the research background and significance of this study. [Section 2](#) introduces the proposed electrochemical degradation model based on SEI layer growth. [Section 3](#) details the framework of the proposed EM-PINN model. [Section 4](#) validates and discusses the performance of the proposed model across different datasets. Finally, [Section 5](#) provides conclusions summarizing the research findings.

## 2. Degradation model based on SEI layer growth

In this subsection, a battery degradation model based on SEI layer growth is established. As shown in [Fig. 1c](#), the battery aging process is described through the explicit equation between SEI layer growth and SOH degradation. Relevant parameter information is provided in [Table 1](#). Finally, physical information from the electrochemical degradation model is embedded into the learning process of the neural network to guide its training, thus enhancing prediction interpretability and improving prediction accuracy.

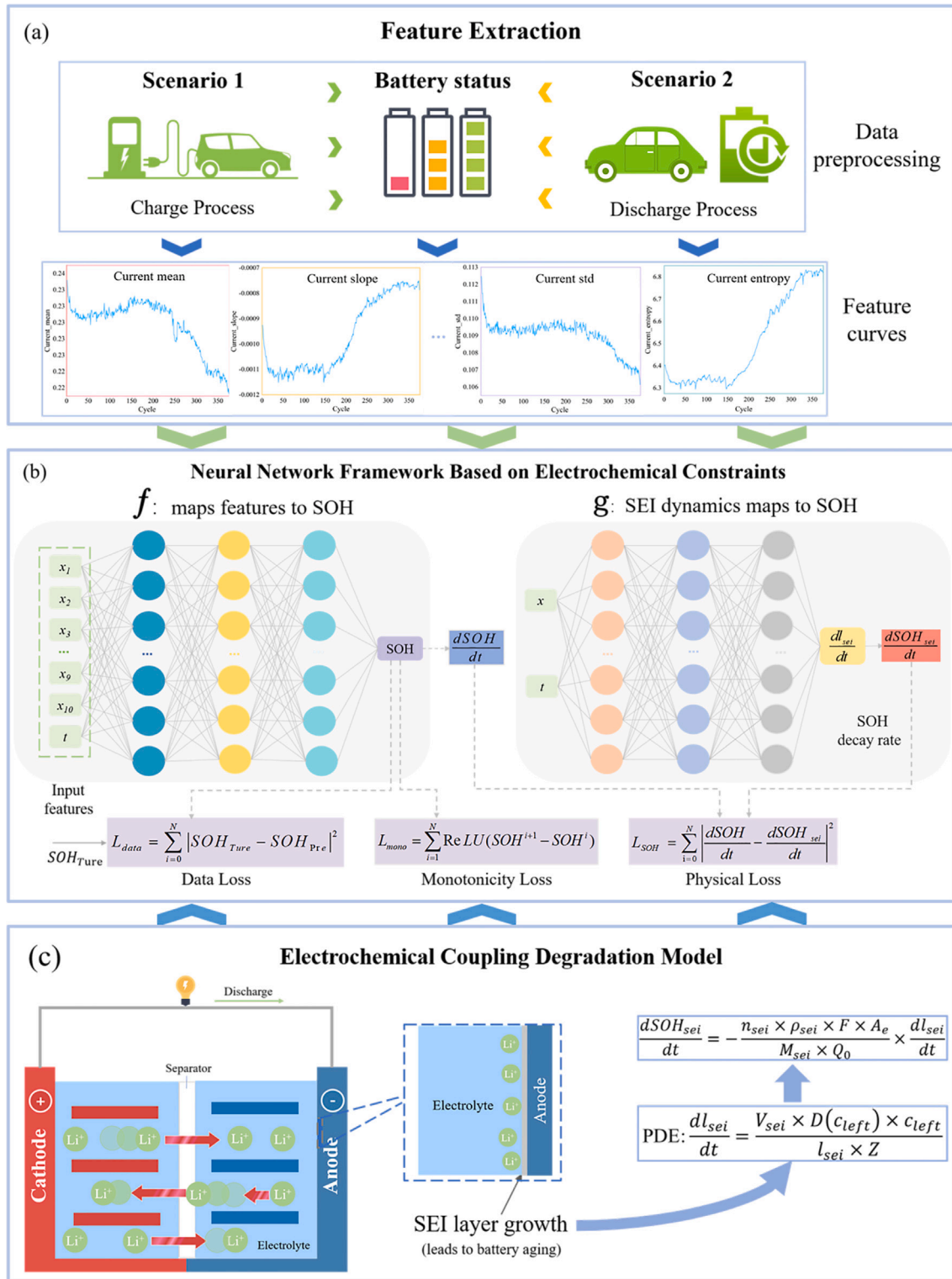


Fig. 1. The framework for estimating battery SOH by integrating the electrochemical model based on SEI layer growth with a neural network model.

### 2.1. SEI layer growth model

The solvent diffusion limited SEI growth model [43] is used in this work. In this model, the growth of the SEI is limited by the diffusion of solvent molecules through the SEI layer. Within the porous SEI, the

solvent is transported via a diffusion process according to:

$$\frac{\partial c}{\partial t} = -\frac{\partial N}{\partial x} \quad (1)$$

where  $c$  represents the solvent concentration,  $x$  denotes the spatial domain, with its value confined to the interval  $[0, L_{sei}]$ . Further details

**Table 1**  
List of parameters of the electrochemical degradation model.

| Parameter    | Description                                                   | Value                    | Ref.                 |
|--------------|---------------------------------------------------------------|--------------------------|----------------------|
| $D_{ref}$    | Reference diffusion coefficient ( $\text{m}^2\text{s}^{-1}$ ) | $1.4523 \times 10^{-13}$ | Tippmann et al. [39] |
| $F$          | Faraday constant ( $\text{C mol}^{-1}$ )                      | 96487                    | Li et al. [23]       |
| $V_{sei}$    | Molar volume of the SEI layer ( $\text{m}^3\text{mol}^{-1}$ ) | $1 \times 10^{-4}$       | Computed             |
| $M_{sei}$    | Molar mass of the SEI layer ( $\text{kg mol}^{-1}$ )          | 0.16                     | [40]                 |
| $\rho_{sei}$ | Density of the SEI layer ( $\text{kgm}^{-3}$ )                | 1600                     | [40]                 |
| $z$          | Electron transfer number for forming 1mol SEI layer           | 2                        | Zhu et al. [41]      |
| $N_{sei}$    | Lithium ions consumed to form 1mol SEI layer                  | 2                        | Zhu et al. [41]      |
| $A_e$        | Electrode surface area ( $\text{m}^2$ )                       | 0.1                      | Tanim et al. [42]    |
| $c_0$        | Initial concentration ( $\text{mol dm}^{-3}$ )                | 1.0                      | Assumed              |
| $L_{sei0}$   | Initial thickness of SEI layer (m)                            | $1 \times 10^{-9}$       | Assumed              |
| $k$          | Reaction rate constant ( $\text{ms}^{-1}$ )                   | $2 \times 10^{-11}$      | Li et al. [23]       |

regarding the spatial domain  $x$  can be found in Supplementary Note 1.  $N$  is the solvent flux. Considering the SEI layer is porous, the diffusion process occurs between its layers. Thus, the solvent flux is defined as:

$$N = -D(c) \frac{\partial c}{\partial x} \quad (2)$$

where  $D(c)$  is the solvent diffusion coefficient. Due to the thickening of the SEI layer and the consumption of solvent concentration, which restrict the diffusion process, it is defined as:

$$D(c) = c \cdot D_{ref} \quad (3)$$

where  $D_{ref}$  is the reference diffusion coefficient.

Since the current density for SEI layer formation is generated by local electron diffusion and referring to Ref. [44], the exchange current density ( $j_{sei}$ ) is defined as:

$$j_{sei} = \frac{-D(c_{left}) \cdot c_{left} \cdot F}{L_{sei}} \quad (4)$$

where  $F$  is the Faraday constant,  $c_{left}$  is defined as the left boundary value of  $c$ . However, further details regarding the specification of boundary conditions can be found in Supplementary Note 2.

Research has shown that the formation of SEI is an irreversible reduction process [36,37], Based on this process and combined with Eq. (4), the thickness growth rate of the SEI layer is derived as:

$$\frac{dL_{sei}}{dt} = -\frac{V_{sei} \cdot j_{sei}}{F \cdot z} = \frac{V_{sei} \cdot D(c_{left}) \cdot c_{left}}{L_{sei} \cdot z} \quad (5)$$

where  $V_{sei}$  is the molar volume of the SEI layer,  $z$  is the electron transfer number for the SEI formation reaction.

## 2.2. Degradation model

By combining Eq. (5) with the established relationship between capacity fade rate and SEI volume growth rate [26], the mathematical relationship between capacity fade rate and SEI layer growth rate is derived as:

$$\frac{dQ}{dt} = -\frac{n_{sei} \cdot \rho_{sei} \cdot F \cdot A_e}{M_{sei}} \frac{dL_{sei}}{dt} \quad (6)$$

where  $n_{sei}$  is the lithium ions consumed to form 1 mol of SEI layer,  $\rho_{sei}$  is the density of the SEI layer,  $A_e$  is electrode area,  $M_{sei}$  is the molar mass of the SEI layer.

The SOH of the battery is defined as the ratio of the present available capacity to the initial capacity [13]:

$$SOH = \frac{Q_{Current}}{Q_{Initial}} \quad (7)$$

Therefore, by combining Eq. (5), Eq. (6) and Eq. (7), the explicit equation between the SEI layer growth rate and the SOH degradation rate is given by:

$$\frac{dSOH_{sei}}{dt} = -\frac{n_{sei} \cdot \rho_{sei} \cdot F \cdot A_e}{M_{sei} \cdot Q_{Initial}} \frac{dL_{sei}}{dt} \quad (8)$$

where Eq. (8) is employed to describe the battery degradation process induced by SEI layer growth and to construct the physical loss term.

Specifically, the attributes affecting SOH degradation are modeled from the perspective of electrochemical equations based on SEI film growth, with a neural network representing function  $g$  being utilized to estimate the SOH degradation rate. The output of this network is subsequently combined with the SOH degradation rate predicted by the neural network for the function  $f$ , thus forming a physical loss term that guides the model training process. The detailed implementation of this procedure is illustrated in Fig. 1(b).

## 3. Physics-informed loss-based PINN framework for SOH estimation

As illustrated in Fig. 1, the EM-PINN framework consists of three key components. First, ten distinct feature variables along with the time series are extracted from the voltage and current curves recorded during charge/discharge cycles to serve as inputs to the neural network, as shown in Fig. 1(a).

Second, a dual-branch PINN framework is designed, as illustrated in Fig. 1(b). In this architecture, one branch is responsible for mapping the extracted feature parameters to the SOH model, where the inputs to this branch network are feature variables  $x_i (i = 1, 2, \dots, 10)$  and  $t$ , with the output being the predicted SOH value. The other branch maps the SEI layer growth kinetics to the SOH degradation process, where the obtained SOH decay rate is incorporated as part of the physical constraint term into the neural network training process. The inputs to this network are  $t$  and  $x$ , while the output is the SOH decay rate.

Finally, based on the mathematical relationship between SEI layer growth and capacity loss, we develop a growth model for the SEI layer to characterize the battery aging process, as shown in Fig. 1(c). This model establishes an explicit correlation between SEI layer growth and SOH degradation, which is subsequently transformed into the physical constraints used in the PINN framework.

In this study, the design of the loss function constitutes a critical component of the overall model. To ensure that the SOH degradation rate generated by the  $g$  module network adhere to the physical principles governing battery degradation, we introduce three dedicated loss terms: the PDE loss, boundary condition loss, and initial condition loss, with their specific expressions provided below:

$$\begin{cases} Loss_{PDE} = Loss_{L_{sei}} + Loss_c \\ Loss_{Init} = Loss_{L_{sei0}} + Loss_{c_0} \\ Loss_{Bound} = Loss_{left} + Loss_{right} \end{cases} \quad (9)$$

where  $loss_{PDE}$ ,  $loss_{Init}$  and  $loss_{Bound}$  represent the residual loss of the partial differential equation, the loss associated with the initial conditions, and the boundary loss, correspondingly.

According to Eq. (9), the three primary loss terms comprise six distinct loss components, with their specific expressions provided below:

$$Loss_{L_{sei}} = \sum_{i=0}^N \left| \frac{dL_{sei}}{dt} - \frac{V_{sei} \cdot D(c_{left}) \cdot c_{left}}{L_{sei} \cdot z} \right|^2 \quad (10)$$

$$Loss_c = \sum_{i=0}^N \left| \frac{dc}{dt} - \frac{L_{sei}'}{L_{sei}} \cdot \xi \cdot \frac{dc}{d\xi} + \frac{1}{L_{sei}} \cdot \frac{dN}{d\xi} \right|^2 \quad (11)$$

where Eq. (10) and Eq. (11) represent the PDE loss terms for SEI layer growth and concentration variation respectively, which collectively constitute the electrochemical model's PDE loss term designed to ensure the SOH degradation rate calculated from the SEI layer satisfies the governing electrochemical equations. Furthermore,  $\xi$  is a scaling factor introduced to address the moving boundary problem, with its values confined to the interval [0, 1]. Further details regarding  $\xi$  can be found in Supplementary Note 1.

$$Loss_{L_{sei0}} = \sum_{i=0}^N |L_{sei} - L_0|^2 \quad (12)$$

$$Loss_{c_0} = \sum_{i=0}^N |c - c_0|^2 \quad (13)$$

where Eq. (12) and Eq. (13) represent the initial condition loss terms for SEI layer thickness and concentration respectively, which are designed to ensure compliance with prespecified conditions at the initial time.

$$Loss_{right} = \sum_{i=0}^N |c - c_0|^2 \quad (14)$$

$$Loss_{left} = \sum_{i=0}^N \left| \frac{dc}{d\xi} - \frac{k \cdot c_{left} \cdot L_{sei}}{D(c_{left})} \right|^2 \quad (15)$$

where Eq. (14) and Eq. (15) represent the boundary condition loss terms for concentration  $c$  at the left and right boundaries respectively, which are designed to ensure that both the concentration and its rate of change conform to the prescribed boundary values under boundary conditions.

However, in electrochemical modeling, the utilization of PINNs for solving partial differential equations presents certain challenges. To address this issue, a simplified approach is adopted without loss of generality. Instead of explicitly solving for the thickness of the SEI layer or the SOH, the rate of SOH degradation induced by SEI growth is derived under the constraints of each loss term in Eq. (9), leveraging the growth rate of the SEI layer. This resulting SOH degradation rate, derived from the electrochemical model, is then incorporated as a key component into the loss function of the subsequent SOH estimation model, thereby overcoming the difficulties associated with directly solving for SOH. It is noteworthy that certain parameters in this model, such as the initial capacity, are calibrated based on the specific capacity of each dataset.

In this study, the neural network framework consists of three types of neural network layers: the input layer, the hidden layer, and the output layer. First, ten features extracted from the voltage–current curves, along with the time series, are introduced into the neural network through the input layer. These inputs are then computed and forward-propagated within the hidden layers. The output layer ultimately generates the predicted battery SOH values. These predictions are compared with the corresponding ground-truth measurements to compute the data loss term, as defined in Eq. (16).

$$Loss_{data} = \sum_{i=0}^N |SOH_{Ture} - SOH_{Pre}|^2 \quad (16)$$

where  $SOH_{Ture}$  denotes the ground-truth value, and  $SOH_{Pre}$  represents the value predicted by the neural network.

Without loss of generality, this study treats battery SOH degradation as a function of time. Consequently, the temporal derivative of the SOH values output by the neural network is computed to obtain the SOH decay rate. This derived rate is then employed to construct a physics-informed residual loss term, which is subsequently compared with the SOH change rate obtained from the physical model. The specific formulation is presented in Eq. (17) and illustrated in Fig. 1(b).

$$Loss_{SOH} = \sum_{i=0}^N \left| \frac{dSOH}{dt} - \frac{dSOH_{sei}}{dt} \right|^2 \quad (17)$$

In addition to the aforementioned SOH loss function, a loss term for irreversible decay [33] is introduced to ensure the neural network's SOH predictions adhere to the physical characteristics of battery degradation. The specific formulation is as follows:

$$Loss_{mono} = \sum_{i=0}^N ReLU(SOH^{i+1} - SOH^i) \quad (18)$$

where  $SOH^{i+1}$  represents the estimated SOH value at the next time step, and  $SOH^i$  corresponds to the estimated SOH value at the current time step.

The overall loss function is defined in Eq. (19).

$$\begin{cases} Loss_{phy1} = \alpha Loss_{SOH} + \beta Loss_{mono} \\ Loss_{phy2} = \gamma Loss_{PDE} + \delta Loss_{Init} + \varepsilon Loss_{Boun} \\ Loss_{Total} = Loss_{data} + Loss_{phy1} + Loss_{phy2} \end{cases} \quad (19)$$

where  $loss_{phy1}$  and  $loss_{phy2}$  represent two distinct terms of physical loss, respectively:  $loss_{phy1}$  comprises the physical loss of SOH and the loss of monotonicity of SOH, while  $loss_{phy2}$  consists of various terms of physical loss from the electrochemical model, as shown in Eq. (9), the  $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\delta$  and  $\varepsilon$  are trade-off parameters. However, further details regarding these weighting parameters are provided in Supplementary Note 3. This approach to loss function construction not only facilitates faster model convergence but also enhances the physical interpretability of the model. Corresponding detailed analysis can be found in Supplementary Note 4.

## 4. Results and discussion

### 4.1. Selection of dataset

To simulate realistic charging and discharging conditions as comprehensively as possible and cover different operational strategies, this study uses data from Batch 1 (8 cells subjected to constant-current (CC) charging at a 2C rate) and Batch 4 (8 cells subjected to dynamic charging/discharging profiles) of the XJTU battery dataset [33]. These batches are denoted as Cell-1 and Cell-4. The capacity degradation trajectories of Cell-1 and Cell-4 over cycle number are depicted in Fig. 2a and Fig. 2b, respectively.

The batteries used in this data set are 18,650 type nickel cobalt manganese (NCM) LIBS with the chemical composition  $LiNi_{0.5}Co_{0.2}Mn_{0.3}O_2$ . These batteries possess a nominal capacity of 2000 mAh, a nominal voltage of 3.6 V, charge and discharge cutoff voltages of 4.2 V and 2.5 V, respectively. All experiments were carried out at room temperature until the batteries were cycled to failure (defined as a capacity drop below 80% of the initial value).

#### 4.1.1. Data preprocessing

Considering potential outliers in the raw data caused by sensor malfunctions or measurement errors, data cleaning was first performed by removing values outside the  $3\sigma$  range to prevent their impact on model prediction accuracy and stability.

To address scale variations among raw features, Min–Max normalization was applied to eliminate dimensional discrepancies between the indicators. Each feature sequence was normalized to the range [−1, 1] using Eq. (20):

$$x_{norm} = 2 \times \left( \frac{x - x_{min}}{x_{max} - x_{min}} \right) - 1 \quad (20)$$

where  $x_{norm}$  denotes the normalized value,  $x$  represents the original data value,  $x_{max}$  and  $x_{min}$  correspond to the maximum and minimum values of the feature data, respectively.

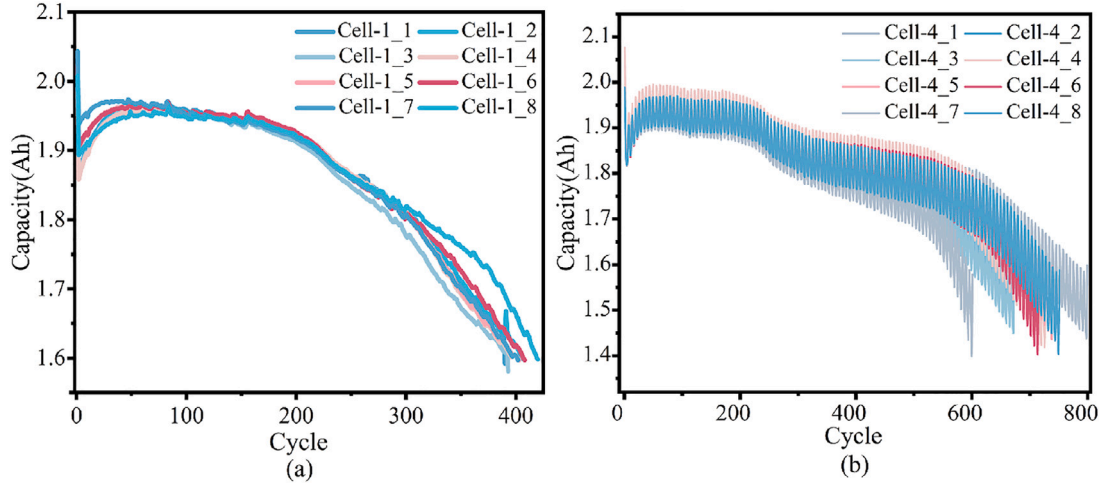


Fig. 2. Capacity degradation trajectories for the Cell-1 and Cell-4 datasets.

**Table 2**  
The Pearson correlation coefficient with the capacity.

| Dataset | Feature 1 | Feature 2 |
|---------|-----------|-----------|
| Batch-2 | −0.85     | −0.85     |
| Batch-3 | −0.14     | −0.15     |
| Batch-5 | 0.43      | 0.43      |
| Batch-6 | 0.39      | 0.37      |

#### 4.1.2. Feature extraction

As feature selection critically influences the accuracy and stability of model predictions, this study conducted a correlation analysis based on feature variables commonly used in the literature [33,45]. The aim was to identify features exhibiting strong correlations with capacity for constructing a stable and accurate EM-PINN model for battery SOH prediction. Following Pearson correlation analysis, the following ten feature variables, along with the time series, were ultimately selected as the final model inputs: capacity during the CC and CV phases, charging time during the CC and CV phases, current mean, current standard deviation, current kurtosis, current skewness, current slope, and current entropy (These features are labeled sequentially 1 to 10 for subsequent identification in correlation analysis).

As shown in Fig. 3, the correlation coefficients between the ten feature variables and capacity are presented for two battery groups under different discharge protocols.

Regarding the phenomenon that Features 1 and 2 exhibit inconsistent correlations across different battery datasets, we preliminarily conjecture that this discrepancy may be attributed to variations in charge/discharge protocols, given that the chemical compositions of these two battery datasets are identical, and the data were collected solely under different charge/discharge protocols.

A controlled variable approach was used to validate this hypothesis. This involved conducting a correlation analysis on datasets from lithium-ion battery groups that shared an identical chemical composition but were subjected to different charge/discharge protocols. The results of this analysis are presented below:

Based on the correlation results in Table 2 and the background that the four datasets were measured under different charge/discharge strategies [33], it can be concluded that Feature 1 and Feature 2 are indeed influenced by different charge/discharge strategies, leading to inconsistent correlation issues.

#### 4.2. Selection of metrics

To scientifically and accurately describe experimental results, this study employs two metrics: MAPE and Root Mean Square Error (RMSE) to compare the prediction performance of several models. The calculation formulas are defined as:

$$\begin{cases} \text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=0}^N |y - \hat{y}|^2} \\ \text{MAPE} = \frac{1}{N} \sum_{i=0}^N \left| \frac{y - \hat{y}}{y} \right| \times 100\% \end{cases} \quad (21)$$

where  $y$  represents the true value and  $\hat{y}$  represents the predicted value of the model.

#### 4.3. Validation on Cell-1 and Cell-4 battery dataset

Following related studies by Ye et al. [31] and Wang et al. [33], this section selects Recurrent Neural Network (RNN), Convolutional Neural Network (CNN), LSTM, and the empirical model-based PINN developed by Wang et al. [33] as benchmark models for comparative testing, in order to validate the performance of the proposed EM-PINN model.

This study primarily focuses on the first and four battery group datasets from the XJTU battery dataset for the discussion of experimental results. Furthermore, For each dataset, the batteries were divided into training data, validation data, and test data sets in an approximate ratio of 6:2:2. During model training, physical information from the electrochemical coupled degradation model in Section 2.1 is incorporated as physical constraints to assist neural network training.

In these experiments, all models were trained using the PyTorch 1.7.1 framework on a processor of 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40 GHz. The proposed prediction module comprises four hidden layers with the following architecture: two layers of 60 neurons and two layers of 32 neurons. For more details regarding the model, please see Supplementary Note 3. The Adam optimizer (initial learning rate: 0.002, dropout rate: 0.2) was uniformly applied to ensure training stability.

The estimation results of the five models (EM-PINN, RNN, CNN, LSTM and PINN) on Cell-1 and Cell-4 datasets are shown in Fig. 4. The experimental results show that the EM-PINN model yields mean MAPE values of 0.73% and 0.83% with corresponding mean RMSE values of 1.00% and 1.15% across the two dataset groups, respectively. In contrast, the RNN, CNN, and LSTM models exhibit significantly higher MAPE and RMSE values than EM-PINN on both datasets. Furthermore, although the PINN model achieves a slightly lower mean MAPE (0.81%) than EM-PINN (0.83%) on the individual Cell-4 dataset,

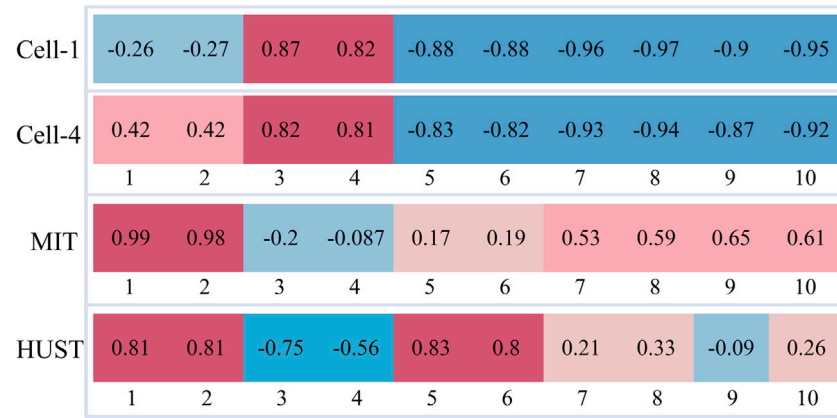


Fig. 3. Correlation heatmap between the ten extracted features and capacity across four datasets.

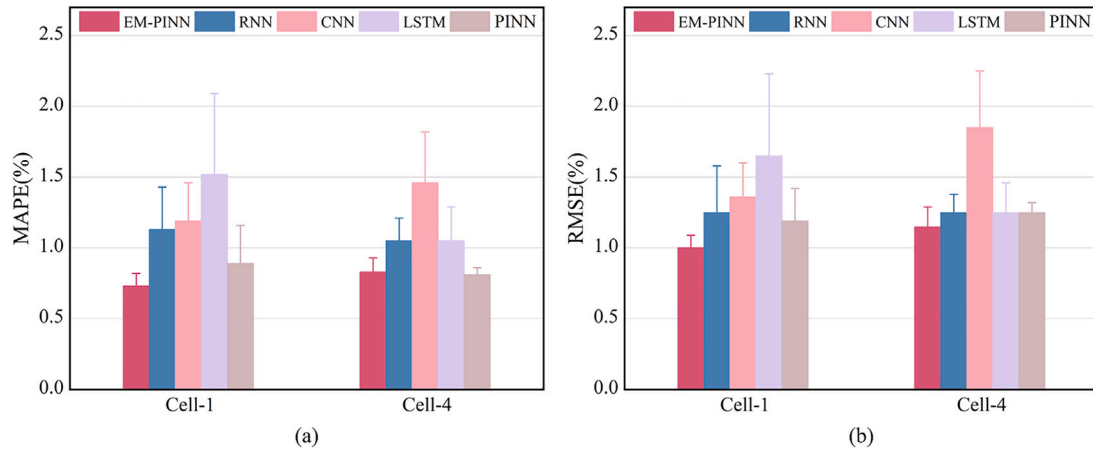


Fig. 4. Estimation errors of five models (EM-PINN, RNN, CNN, LSTM and PINN) for Cell-1 and Cell-4 datasets: (a) MAPE for prediction results of the five models; (b) RMSE for prediction results of the five models. In the histogram, the bars represent the mean values of each group, and the error bars indicate the corresponding standard deviations.

Table 3

Estimation errors of EM-PINN, RNN, CNN, LSTM and PINN over 9 experimental trials on Cell-1 and Cell-4 datasets.

| Experiment | EM-PINN |         | RNN     |         | CNN     |         | LSTM    |         | PINN    |         |
|------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
|            | MAPE    | RMSE    | MAPE    | RMSE    | MAPE    | RMSE    | MAPE    | RMSE    | MAPE    | RMSE    |
| 1          | 0.00793 | 0.00976 | 0.01758 | 0.01996 | 0.01429 | 0.01574 | 0.02010 | 0.02048 | 0.01623 | 0.01752 |
| 2          | 0.00812 | 0.01003 | 0.00837 | 0.00930 | 0.01452 | 0.01507 | 0.00778 | 0.00910 | 0.00807 | 0.01199 |
| 3          | 0.00656 | 0.01038 | 0.01200 | 0.01353 | 0.00877 | 0.01197 | 0.01792 | 0.01938 | 0.00829 | 0.01164 |
| 4          | 0.00595 | 0.00949 | 0.00885 | 0.01031 | 0.01099 | 0.01196 | 0.01664 | 0.01876 | 0.00798 | 0.01238 |
| 5          | 0.00778 | 0.00934 | 0.01175 | 0.01235 | 0.00661 | 0.00864 | 0.01194 | 0.01256 | 0.00729 | 0.00982 |
| 6          | 0.00671 | 0.00912 | 0.01024 | 0.01089 | 0.01472 | 0.01604 | 0.01365 | 0.01457 | 0.00763 | 0.00952 |
| 7          | 0.00854 | 0.01068 | 0.00804 | 0.00933 | 0.01408 | 0.01607 | 0.02241 | 0.02369 | 0.00761 | 0.00999 |
| 8          | 0.00673 | 0.00953 | 0.01358 | 0.01523 | 0.01212 | 0.01418 | 0.02273 | 0.02474 | 0.00821 | 0.01178 |
| 9          | 0.00659 | 0.00953 | 0.01322 | 0.01370 | 0.01240 | 0.01329 | 0.01198 | 0.01311 | 0.01006 | 0.01320 |

its RMSE and corresponding standard deviations are higher, indicating inferior stability compared to EM-PINN. Additionally, EM-PINN demonstrates a 0.7% lower overall MAPE than the PINN model across both datasets.

To comprehensively demonstrate model performance while avoiding coincidental outcomes and eliminating the impact of training randomness, 9 repeated experimental trials were conducted for five models on both the Cell-1 and Cell-4 datasets. The prediction curves were plotted based on the predicted versus true values from each trial, and

the MAPE/RMSE values were statistically analyzed in 9 experiments, as shown in Figs. 5,6 and Table 3.

Based on the analysis of error metrics from 9 experimental trials, it can be observed that although the prediction errors of the other four models are slightly lower than those of EM-PINN in a small number of cases, the proposed EM-PINN achieves not only the lowest MAPE and RMSE in the majority of experiments but also consistently exhibits the smallest fluctuation. Furthermore, both the optimal and average predicted values of EM-PINN exceed those of the other four models

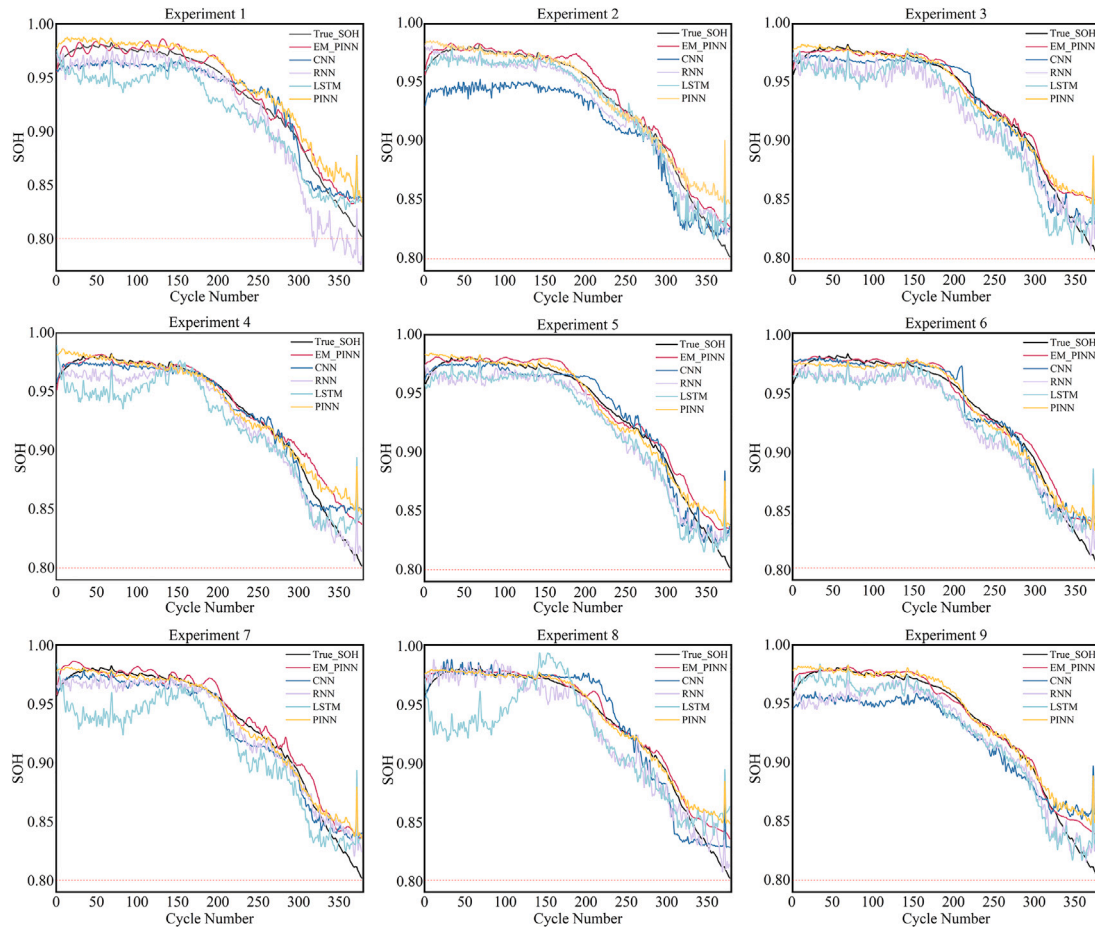


Fig. 5. Predicted versus measured SOH curves obtained from 9 experimental trials of the five models on the Cell-1 dataset.

**Table 4**

Estimation Errors of the Two Models (EM-PINN, PINN) on Cell-1 and Cell-4 Datasets. The reported experimental results represent the average values obtained from ten independent experimental trials.

| Dataset | EM-PINN       |               | PINN   |        |
|---------|---------------|---------------|--------|--------|
|         | MAPE          | RMSE          | MAPE   | RMSE   |
| Cell-1  | <b>0.0247</b> | <b>0.0389</b> | 0.0326 | 0.0453 |
| Cell-4  | <b>0.0191</b> | <b>0.0377</b> | 0.0263 | 0.0443 |

(RNN, CNN, LSTM and PINN). These results further confirm that the proposed EM-PINN model achieves high prediction accuracy while maintaining superior stability throughout the SOH prediction process.

Meanwhile, the prediction curves from 9 experimental trials clearly demonstrate that the proposed model maintains the most stable trajectory across different battery datasets (Cell-1 and Cell-4). In contrast, the prediction curves of the other four models exhibit significantly greater fluctuations, indicating their higher sensitivity to variations in data distribution and limited generalization capability—particularly in the case of the CNN model.

Furthermore, to validate the early-prediction capability of the proposed model, this experiment utilized partial cycling data from the Cell-1 and Cell-4 datasets. Specifically, the model was trained only on a portion of the cycling data and was then required to predict the complete SOH degradation trajectory until end-of-life. In this experiment, the first 300 cycles (corresponding to a capacity fade to approximately 80% of the initial value) of six cells (serial numbers 1, 2, 3, 5, 6, 7) from the Cell-1 dataset and the first 500 cycles (corresponding to a

capacity fade to approximately 85% of the initial value) of the same six cells from the Cell-4 dataset were employed as the training set, while two cells with complete cycling records were used as the test set. To assess the accuracy of the model, the PINN framework proposed by Wang et al. [33] was adopted as a benchmark for comparison. After complete training, the prediction results of the two models on the two different datasets are presented in Table 4. It can be observed that, although the performance of the proposed model decreases under this scenario, the proposed EM-PINN model still outperforms the PINN model. This outcome demonstrates that the proposed PINN model incorporating electrochemical mechanisms exhibits better stability than the empirically-based PINN model.

#### 4.4. Ablation study

To evaluate the impact of the embedded physics-based loss terms from the electrochemical model, this section presents an ablation study focusing on the SOH physical loss term and the electrochemical model-derived loss components, aiming to validate the necessity of incorporating these elements. The study utilizes datasets from Cell-1 and Cell-4 as test cases, with detailed descriptions of the datasets provided in Section 4.1.

In this ablation study, the model retaining only the data loss and irreversible decay loss terms is designated as the benchmark for comparison, named “NN”. Except for the absence of physics-based loss terms, this reference model maintains identical configuration parameters and network architecture to the EM-PINN framework. Subsequently, comparative training and evaluation between the “NN” and EM-PINN models are conducted to validate the effectiveness and necessity of

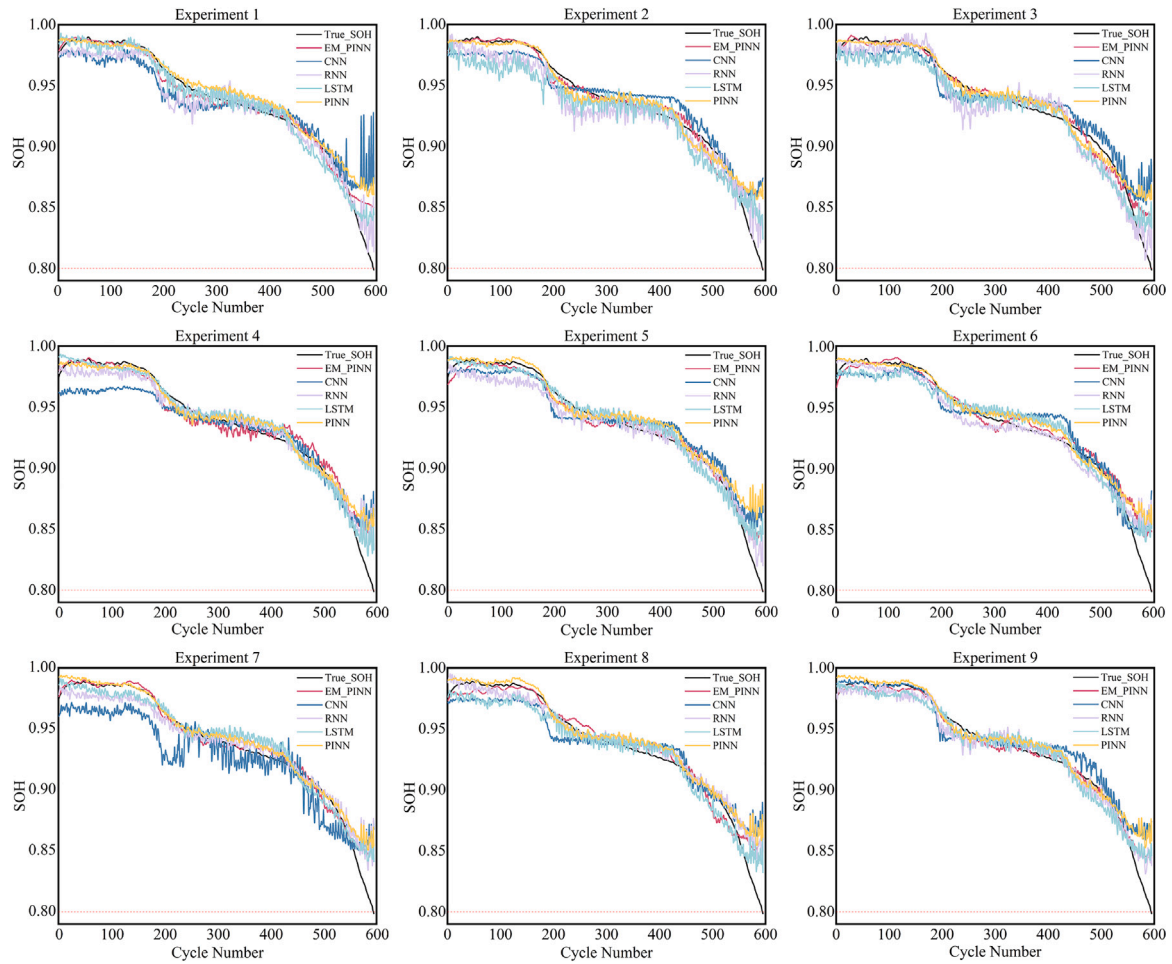


Fig. 6. Predicted versus measured SOH curves obtained from 9 experimental trials of the five models on the Cell-4 dataset.

**Table 5**

Estimation errors of the two models (EM-PINN, NN) on Cell-1 and Cell-4 datasets.

| Dataset | EM-PINN        |                | NN      |         |
|---------|----------------|----------------|---------|---------|
|         | MAPE           | RMSE           | MAPE    | RMSE    |
| Cell-1  | <b>0.00808</b> | <b>0.01010</b> | 0.01136 | 0.01399 |
| Cell-4  | <b>0.00632</b> | <b>0.00924</b> | 0.01061 | 0.01321 |

incorporating physics-based loss components, using the two metrics introduced in Section 4.2: RMSE and MAPE.

The experimental results obtained from the model predictions after the complete training procedure are summarized in Table 5. Analysis of the data shows that the EM-PINN model, which incorporates physics-informed loss terms, outperforms the “NN” benchmark model that lacks these physical constraints in terms of both predictive accuracy and stability. Specifically, the EM-PINN model achieves average reductions of 34.6% in MAPE and 28.9% in RMSE.

Furthermore, to comprehensively and clearly visualize the prediction outcomes and comparative performance of the different models, the prediction curves of both models are plotted and presented in Fig. 7. The graphical results demonstrate that the EM-PINN prediction curve exhibits closer alignment with the ground-truth trajectory and shows significantly reduced fluctuation compared to the “NN” prediction profile. This provides compelling evidence that incorporating physics-informed loss terms enhances the model’s predictive capability and stability.

#### 4.5. Validation on MIT battery dataset

This section presents the experimental results obtained by selecting 10% (10 training batteries, 2 testing batteries) and 20% (20 training batteries, 4 testing batteries) of the samples from the MIT battery dataset [46] as the test subjects under small-sample scenarios. Consistent with Section 4.3, we select RNN, CNN, LSTM and PINN as baseline models for comparison. This dataset comprises 124 lithium iron phosphate (LFP)/graphite cylindrical cells (model APR18650M1 A) with a nominal capacity of 1.1 Ah and a nominal voltage of 3.3 V. All batteries were discharged at a 4C rate and charged using either one-step or two-step fast-charging protocols. CC charging terminated when the batteries reached 80% state of charge (SOC), followed by CC-CV charging at a 1C rate. The charge cutoff voltage and discharge cutoff voltage were set at 3.6 V and 2.0 V, respectively.

In the experiments, the dataset is divided into training batteries and test batteries in an 8:2 ratio, which differs from the partitioning approach applied to the XJTU dataset in Section 4.1. Secondly, to ensure consistency in the data sequences used by each model and to avoid the impact of subjective sample selection on the experimental results, the same random seed was set during the training process of each model. Through separate training and testing of the five models, the validation error results are shown in Fig. 8. It can be observed from the figure that the proposed EM-PINN model achieves the lowest and most stable values on both metrics. Specifically, with 10% of the data samples, the MAPE and RMSE are 0.873% and 0.925%, respectively; with 20% of the data samples, the MAPE and RMSE are 0.722% and 0.984%, respectively.

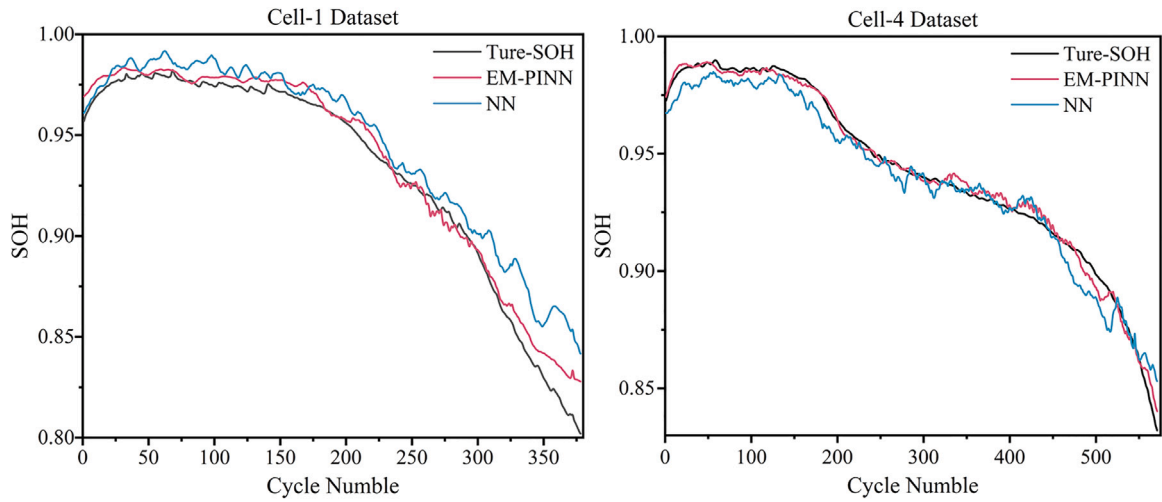


Fig. 7. Estimation Errors of the two Models (EM-PINN, NN) on Cell-1 and Cell-4 Datasets.

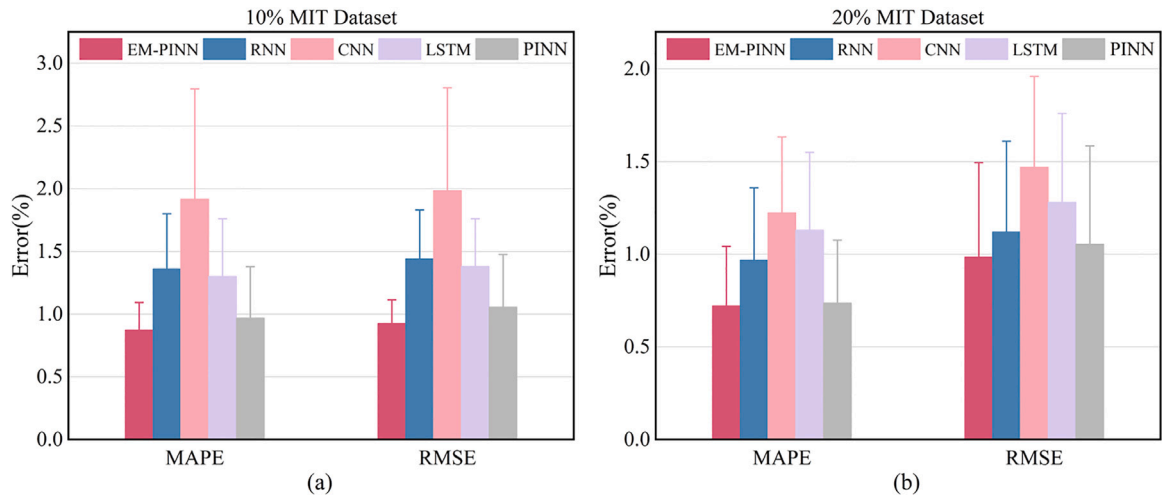


Fig. 8. The estimation errors of the five models on the MIT dataset, comprising the metrics MAPE and RMSE.

Table 6

Estimation errors of EM-PINN, RNN, CNN, LSTM and PINN over 9 experimental trials on 10% MIT datasets.

| Experiment | EM-PINN |         | RNN     |          | CNN     |         | LSTM    |         | PINN    |         |
|------------|---------|---------|---------|----------|---------|---------|---------|---------|---------|---------|
|            | MAPE    | RMSE    | MAPE    | RMSE     | MAPE    | RMSE    | MAPE    | RMSE    | MAPE    | RMSE    |
| 1          | 0.01105 | 0.01140 | 0.01853 | 0.019802 | 0.02528 | 0.02634 | 0.01225 | 0.01291 | 0.01999 | 0.02122 |
| 2          | 0.00883 | 0.00954 | 0.01259 | 0.015027 | 0.01994 | 0.02250 | 0.01154 | 0.01216 | 0.00891 | 0.00965 |
| 3          | 0.00890 | 0.00958 | 0.01686 | 0.017878 | 0.02964 | 0.02929 | 0.0196  | 0.01948 | 0.01036 | 0.01078 |
| 4          | 0.00640 | 0.00674 | 0.01297 | 0.013058 | 0.03733 | 0.03616 | 0.01098 | 0.01180 | 0.01082 | 0.01309 |
| 5          | 0.00845 | 0.00899 | 0.01286 | 0.014929 | 0.01356 | 0.01509 | 0.00780 | 0.00965 | 0.00866 | 0.00919 |
| 6          | 0.00821 | 0.00892 | 0.01190 | 0.012047 | 0.01038 | 0.01227 | 0.01102 | 0.01264 | 0.00884 | 0.00932 |
| 7          | 0.00851 | 0.00889 | 0.01186 | 0.011910 | 0.01324 | 0.01391 | 0.01197 | 0.01301 | 0.00787 | 0.00854 |
| 8          | 0.01013 | 0.01065 | 0.01502 | 0.014885 | 0.01324 | 0.01338 | 0.01824 | 0.01840 | 0.00841 | 0.00913 |
| 9          | 0.00765 | 0.00798 | 0.01287 | 0.012898 | 0.01321 | 0.01396 | 0.01567 | 0.01589 | 0.00782 | 0.00828 |

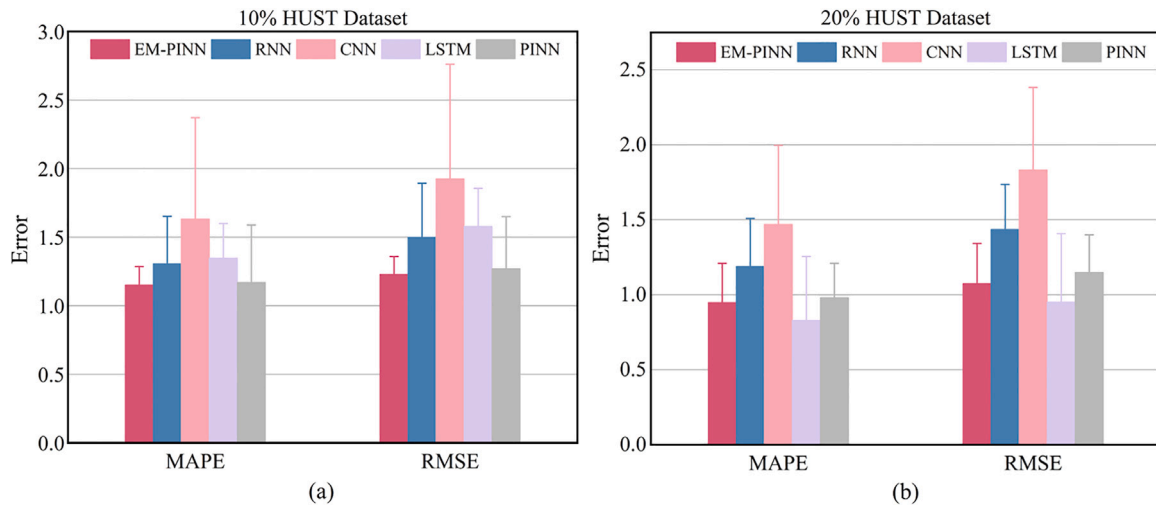
Furthermore, the error results from nine experiments for each of the five models under two different sample sizes were statistically evaluated to mitigate the effects of randomness. The results of these experiments are presented in Table 6 and Table 7, respectively. Although benchmark models occasionally achieve marginally lower prediction errors than the proposed EM-PINN model in isolated experiments, they

exhibit significant fluctuations across different experimental trials. In contrast, the EM-PINN model demonstrates superior performance with lower prediction errors than the other four models in the vast majority of cases, while simultaneously showing the smallest variations across multiple tests. These results further validate the reliability and accuracy of the proposed approach.

**Table 7**

Estimation errors of EM-PINN, RNN, CNN, LSTM and PINN over 9 experimental trials on 20% MIT datasets.

| Experiment | EM-PINN |         | RNN     |         | CNN     |         | LSTM    |         | PINN    |          |
|------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|----------|
|            | MAPE    | RMSE    | MAPE    | RMSE    | MAPE    | RMSE    | MAPE    | RMSE    | MAPE    | RMSE     |
| 1          | 0.00675 | 0.00938 | 0.00930 | 0.01148 | 0.01107 | 0.01353 | 0.01096 | 0.01195 | 0.00688 | 0.01077  |
| 2          | 0.00777 | 0.01063 | 0.00826 | 0.00956 | 0.01187 | 0.01436 | 0.01120 | 0.01274 | 0.00881 | 0.01296  |
| 3          | 0.00705 | 0.00950 | 0.00944 | 0.01259 | 0.01162 | 0.01397 | 0.01088 | 0.01253 | 0.00719 | 0.01136  |
| 4          | 0.00775 | 0.01036 | 0.01086 | 0.01174 | 0.01118 | 0.01386 | 0.01128 | 0.01316 | 0.00723 | 0.01032  |
| 5          | 0.00694 | 0.00935 | 0.01102 | 0.01292 | 0.01160 | 0.01335 | 0.01158 | 0.01296 | 0.00596 | 0.00913  |
| 6          | 0.00726 | 0.00956 | 0.01004 | 0.01030 | 0.01697 | 0.01991 | 0.00955 | 0.01155 | 0.00756 | 0.01016  |
| 7          | 0.00709 | 0.01025 | 0.00888 | 0.00982 | 0.01237 | 0.01479 | 0.01193 | 0.01331 | 0.00820 | 0.01109  |
| 8          | 0.00661 | 0.00935 | 0.00890 | 0.01083 | 0.01226 | 0.01482 | 0.01008 | 0.01175 | 0.00710 | 0.00966  |
| 9          | 0.00648 | 0.00907 | 0.01061 | 0.01112 | 0.00981 | 0.01249 | 0.01631 | 0.01631 | 0.00721 | 0.009852 |

**Fig. 9.** Estimation errors of five models (EM-PINN, RNN, CNN, LSTM and PINN) for HUST Dataset.

Overall, under small-sample conditions, the proposed EM-PINN model, with the incorporation of physical constraints from electrochemical mechanisms, shows steady performance improvement. It effectively reduces the dependency on large datasets to a certain extent while maintaining high prediction accuracy.

#### 4.6. Validation on HUST battery dataset

To further validate the generality and stability of the EM-PINN model, the HUST dataset [47] was used to evaluate its performance.

A total of 77 lithium iron phosphate/graphite batteries with a nominal capacity of 1.1 Ah and nominal voltage of 3.3 V are included in this dataset. Under constant 30 °C conditions, these batteries underwent cyclic testing using identical charging protocols but different multi-stage discharge schemes until failure (SOH below 80%). Consistent with Section 4.3, this section selects RNN, CNN, LSTM and PINN as benchmark models for comparison.

In the experimental, the sample size selection and dataset partitioning methodology were consistent with the approach described in Section 4.5.

The model was evaluated under small-sample conditions using 10% (5 training samples, 2 test samples) and 20% (11 training samples, 4 test samples) of the HUST dataset. As shown in Fig. 9, the proposed model demonstrates exceptional performance with only 10% of the data, achieving MAPE and RMSE values of 1.15% and 1.22%, respectively. The corresponding standard deviations for these metrics are measured at 0.135% and 0.13%, respectively, which are recognized as

the lowest levels among all compared methods. Although comparable prediction accuracy to EM-PINN is shown by the benchmark PINN model, its standard deviations (0.42% and 0.38%) are found to be significantly larger than those of EM-PINN, indicating inferior stability.

When the sample size increases to 20%, slightly lower prediction errors in terms of MAPE and RMSE compared to EM-PINN. However, the lowest standard deviations (0.26% and 0.28%) for these metrics, while the standard deviations of LSTM are observed to be 1.5 times higher. This suggests greater performance variability of the LSTM model across different datasets and confirms the superior stability of EM-PINN. Furthermore, superior performance in both accuracy and stability is demonstrated by EM-PINN when compared to the PINN, CNN, and RNN models.

In conclusion, compared with pure data-driven models, the prediction accuracy and stability of EM-PINN model are improved to a certain extent after integrating physical models under the condition of limited data samples.

## 5. Conclusion

In this study, the limitations of solely physics-based and purely data-driven approaches for predicting battery health states are addressed by introducing an electrochemical model based EM-PINN framework for SOH estimation. First, an electrochemical model describing battery capacity degradation is established based on the degradation mechanism of SEI layer growth. Second, physical information from this electrochemical model is integrated as physical constraints into the

neural network, forming the EM-PINN framework. By incorporating PDE loss, boundary loss, initial condition loss, and SOH degradation rate loss, the physical interpretability of the model is enhanced, improving prediction stability and accuracy.

Finally, to validate the stability and accuracy of our proposed model, we conducted evaluations on datasets under two distinct charge/discharge protocols (Cell-1 and Cell-4), MIT and HUST datasets. The results, compared against those from RNN, CNN LSTM and PINN models, demonstrate that the EM-PINN model consistently achieves superior prediction accuracy and stability across diverse data scenarios. Furthermore, an ablation study was conducted to validate the effectiveness of the physics-informed loss terms incorporated in the EM-PINN model. The experimental results demonstrate a discernible reduction in prediction errors after integrating these physical constraints, thereby providing solid evidence for their efficacy.

In summary, the proposed method exhibits high performance across diverse datasets and operating conditions, offering more reliable technical support for battery health management and lifespan prediction. It should be noted, however, that the current model does not yet incorporate critical factors such as temperature effects on battery performance, which will be a primary focus of our future work.

### CRediT authorship contribution statement

**Kai Gao:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Qi Li:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization. **Lin Hu:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Jing Huang:** Writing – review & editing, Supervision, Project administration, Investigation. **Heng Li:** Writing – review & editing, Supervision, Project administration, Investigation. **Yue Wu:** Writing – review & editing, Supervision, Investigation.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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### Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.energy.2025.139753>.

### Data availability

The data that has been used is confidential.

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